Covering 1024 syndromes with 50 columns

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Abstract

We give a parity-check matrix $\boldsymbol{H}$ of a binary $[50, 40]_2$ code with covering radius 2, so $\ell_2(10, 2) \leq 50$. The best previous length was 51, published by Kaikkonen and Rosendahl in 2003. That length-51 code still seeds the radius-2 family of Davydov, Marcugini, and Pambianco (arXiv:2511.02542). The columns of $\boldsymbol{H}$ admit a $(2, 0)$-partition into $p(\boldsymbol{H}) = 10$ blocks, one block fewer than the partition known for the length-51 code. Construction QM_2^2 therefore applies to $\boldsymbol{H}$. It gives codes of lengths 815 and 1631 at redundancies 18 and 20, both verified by exhaustive syndrome enumeration, and an infinite family with covering density $2601/2048 \approx 1.27002$, against the previous family density $2704/2048 \approx 1.32031$. Every claim reduces to a finite statement about committed matrices. The matrices, the verifiers, and this source are at `https://github.com/wustep/maths`.

1 Definitions and background

1.1 Covering radius

Let $C \leq \mathbb{F}_2^n$ be a binary linear code of codimension r, so $|C| = 2^{n-r}$. The *covering radius* of C is the least R such that every vector of $\mathbb{F}_2^n$ has Hamming distance at most R to some codeword. Let $\boldsymbol{H}$ be an $r \times n$ parity-check matrix of C with column set $S \subset \mathbb{F}_2^r$. Then the covering radius is the least R such that every syndrome is a sum of at most R columns of $\boldsymbol{H}$. Write $\ell_2(r, R)$ for the least length of a binary linear code with codimension r and covering radius R.

A binary linear code of length n and dimension $k = n - r$ is an $[n, k]_2$ code. When we assert the covering radius we append it, as in $[n, k]_2\, R$. For $R = 2$ the covering condition on the column set is

$$\{0\} \cup S \cup (S + S) = \mathbb{F}_2^r, \qquad S + S = \{\, s + s' : s, s' \in S \,\}. \tag{1}$$

A $(2, 0)$-*partition* of the columns (Definition 3.2 of [1]) is a partition into nonempty blocks such that every syndrome, including 0, is a sum of at most two columns from *distinct* blocks. Write $p(\boldsymbol{H})$ for the number of blocks of such a partition.

1.2 Density and Green's problem 40

The covering density of an $[n, n-r]_2\, 2$ code is the exact rational

$$\mu = \frac{1 + n + \binom{n}{2}}{2^r},$$

the number of radius-2 representations divided by the number of syndromes. When (1) holds, $\mu \geq 1$. Green's Problem 40 [4] concerns the asymptotic quantity

$$f(2) = \liminf_{r \to \infty} \mu\big(\ell_2(r, 2), r\big).$$

For an infinite family of $[n(r), n(r)-r]_2\,2$ codes, write $\bar{\mu}(2)$ for the lim inf of μ along the family. The optimal lengths satisfy $\ell_2(r,2) \le n(r)$, so any family bound $\bar{\mu}(2) \le c$ implies $f(2) \le c$. Whether $f(2) = 1$ is open.

The bound $f(2) \le 729/512 \approx 1.42383$ held from 1992 to 2025. Davydov, Marcugini, and Pambianco [1] improved it to $2704/2048 \approx 1.32031$ in November 2025. Their tool is a construction, QM_2^2, that produces an infinite family from one short radius-2 code. The construction applies when the seed admits a $(2,0)$-partition into few enough blocks. Section 2.5 states its exact input and output.

1.3 The seed at $r = 10$

Table 5.1 of [1], row $r = 10$, lists $n = 51$. The entry is the Kaikkonen–Rosendahl code of 2003 [2], of density $1327/1024 \approx 1.29590$. The 2025 paper did not improve it. The infinite family of [1] (Theorem 5.7) has lengths

$$n(r) = 26 \cdot 2^{r/2-4} - 1 = 2^m(51+1) - 1, \qquad m = r/2 - 5.$$

The 51 in this formula is the length-51 code. The whole family is QM_2^2 iterated from that single table entry. Theorem 5.7(i) of [1] calls the 2-partitions of the seed "very important for the next iterative process". Their computer search gives $p(\boldsymbol{H}_{KR}) = 11$ for the length-51 matrix.

A length-50 matrix at $r = 10$ therefore has two effects. It improves one table entry. If it also admits a small $(2,0)$-partition, it re-seeds the whole family. This note gives such a matrix and such a partition.

2 Results

2.1 The code

Theorem 1. *Let $\boldsymbol{H}$ be the 10×50 matrix over $\mathbb{F}_2$ whose columns are the 50 integers of Table 1, where bit i of an integer, counted from the least significant bit, is row $i+1$. Then $\boldsymbol{H}$ has rank 10, its columns are distinct and nonzero, and condition (1) holds. The code with parity-check matrix $\boldsymbol{H}$ is a $[50,40]_2$ code with covering radius exactly 2, and*

$$\ell_2(10,2) \le 50.$$

The proof is a finite computation over the committed matrix. Section 4 describes the two independent programs that perform it and the commands that replay it. Appendix A prints the matrix as bits.

1	2	4	15	16	32	65	86	128	173
183	202	212	247	256	297	320	329	341	366
373	381	391	403	438	460	479	491	502	559
576	608	653	734	742	754	771	777	789	821
846	855	869	881	893	897	927	981	1003	1004

Table 1: The columns of $\boldsymbol{H}$ as LSB-first integers.

The $1276 = 1 + 50 + \binom{50}{2}$ sums of at most two columns hit the 1024 syndromes with multiplicity histogram $\{1{:}859,\ 2{:}129,\ 4{:}24,\ 5{:}9,\ 6{:}3\}$. No multiplicity is zero, which proves (1). Exactly 973 syndromes are neither 0 nor a column, so the covering radius is not 1. The density is $\mu = 1276/1024 = 319/256 = 1.24609375$. The columns are distinct and nonzero, and $\boldsymbol{H}$ has a linearly dependent column triple, so the minimum distance is $d = 3$.

claim	value
shape	10×50, $\mathbb{F}_2$-rank 10, 50 distinct nonzero columns
coverage	1024/1024 syndromes
covering radius	exactly 2
density μ	$319/256 = 1.24609375$
minimum distance	$d = 3$
KR baseline	51 columns, 1024/1024, density 1327/1024

2.2 Minimality

Proposition 2. *Every matrix obtained from* $\boldsymbol{H}$ *by deleting one column violates* (1)*. Each single deletion leaves at least* 9 *of the* 1024 *syndromes uncovered. The minimum* 9 *is attained exactly by the columns* 381*,* 479*, and* 927*.*

In geometric language, the column set of $\boldsymbol{H}$ is a minimal 1-saturating set in $PG(9, 2)$. The covering-code literature calls such a code locally optimal (LO). Minimality concerns this particular 50-set. It says nothing about the existence of a covering 49-set.

2.3 A $(2, 0)$-partition into ten blocks

Proposition 3. *The partition of Table 2 is a* $(2, 0)$*-partition of the columns of* $\boldsymbol{H}$ *into* 10 *blocks.*

block	columns
0	$2, 128, 202, 212, 771, 855, 897, 981$
1	86
2	$381, 893, 1003$
3	$183, 297$
4	$1, 65, 247, 256, 320, 438, 502, 734$
5	$15, 173, 329, 366, 460, 559, 653, 846$
6	$4, 16, 391, 491, 742, 754, 869, 881$
7	$479, 1004$
8	403
9	$32, 341, 373, 576, 608, 777, 789, 821, 927$

Table 2: A $(2, 0)$-partition of the columns of $\boldsymbol{H}$ with $p(\boldsymbol{H}) = 10$.

Each of the 973 syndromes that need a pair has a pair whose two columns lie in distinct blocks. Over those 973 syndromes the pair multiplicities are {1:821, 2:123, 4:19, 5:8, 6:2}. In particular, 821 syndromes have a unique pair, and each unique pair forces its two columns into distinct blocks. The computer search of [1] gives $p(\boldsymbol{H}_{KR}) = 11$ for the length-51 code. We do not claim that 10 blocks is minimal for $\boldsymbol{H}$.

2.4 Dependent triples

$\boldsymbol{H}$ has exactly 10 linearly dependent triples of columns. Exactly one of them, $(491, 734, 821)$, meets three distinct blocks of the partition (blocks 6, 4, and 9). This is the analogue of Theorem 5.2(ii) of [1] and is the extra hypothesis that Construction QM_5^2 needs at $r = 28$. The artifact `result/` does not use it. Later work does (Remark 1).

2.5 Propagation

Construction QM_2^2 (Theorem 4.1 of [1]) takes an $[n_0, n_0 - r_0]_2\, 2$ code whose parity-check matrix $\boldsymbol{H}_0$ has a $(2,0)$-partition into $p(\boldsymbol{H}_0)$ blocks, together with an integer m such that $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$. Its indicator set is $\mathscr{B} = \mathbb{F}_{2^m}$ and its tail block is $\boldsymbol{D} = \boldsymbol{D}_1(2)$. The output is a code with

$$n = 2^m(n_0+1) - 1, \qquad r = r_0 + 2m, \qquad p(\boldsymbol{H}_C) \leq 2^{m+1} + 1. \tag{2}$$

With $n_0 = 50$ and $p(\boldsymbol{H}_0) = 10$ the hypothesis permits exactly $m = 4$ and $m = 5$, since $2^6 = 64 > 50$. We built both outputs from equations (4.2) and (4.4) of [1] and verified them by enumeration of every syndrome:

	r	n	coverage	density	published
seed	10	50	1024/1024	319/256	51 (KR 2003)
$m = 4$	18	815	262144/262144	332521/262144	831
$m = 5$	20	1631	1048576/1048576	1330897/1048576	1663

The indicator assignment inside QM_2^2 is a free choice. A second assignment gives a second pair of matrices, committed under `result/data/alt/`, and both pairs pass the same verification.

2.6 The family

Iterate (2) from $(r,p) = (10,10)$, always with the partition bound $p(\boldsymbol{H}_C) \leq 2^{m+1}+1$. The even $r \leq 64$ reachable by QM_2^2 are

$$10, 18, 20, 30, 32, 34, 36, 38, 40, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64.$$

The even r not reachable this way are $12, 14, 16, 22, 24, 26, 28, 42, 44$. The paper fills its own gaps at $r = 22, 24, 26$ with QM_3^2 and at $r = 28$ with QM_5^2. On the reachable r the lengths and densities have the closed form

$$n = 51 \cdot 2^{r/2-5} - 1, \qquad \mu = \frac{51^2}{2^{11}} - \frac{51}{2^{r/2+1}} + \frac{1}{2^r} \nearrow \frac{2601}{2048} \approx 1.27002.$$

Hence $\bar{\mu}(2) \leq 2601/2048$, and therefore $f(2) \leq 2601/2048$.

	$\bar{\mu}(2)$	source
pre-2025	$729/512 \approx 1.42383$	standing since 1992
arXiv:2511.02542	$2704/2048 \approx 1.32031$	seeded by KR $n_0 = 51$
this seed	$2601/2048 \approx 1.27002$	seeded by $n_0 = 50$

The family rows past $m = 5$ use the bound $p(\boldsymbol{H}_C) \leq 2^{m+1}+1$ of [1], not computed partitions. That is the weakest step in the family, and Section 5 records it.

Remark 1 (later constructions). After the first version of this note, further work applied QM_3^2, QM_5^2, and QM_2^1 to the same seed, together with radius-3 and radius-4 constructions of [1]. The radius-2 outputs at $r = 22, 24, 26, 28$ were verified by exhaustive enumeration, up to all 2^{28} syndromes at $r = 28$. The log `ATTACK.md` and the directory `compute/` record the certified values and the replay scripts. This note is limited to the seed, the partition, and the $m = 4, 5$ outputs.

3 The search

The matrix $\boldsymbol{H}$ is the output of a simulated-annealing local search.[1] The search is not part of the proof. We describe it because the method is simple and transfers to nearby instances.

3.1 Seed

The seed is the Kaikkonen–Rosendahl set: the ten unit vectors and the 41 columns printed in hexadecimal in Theorem 4.3 of [1]. The script reads the hex words LSB-first, while the paper prints them MSB-first. The two readings differ by row reversal, an invertible linear map on $\mathbb{F}_2^{10}$, so no claim depends on the orientation (Section 4.3).

Among the 51 single deletions from this set, deleting the column `0x1A3` $= 419$ leaves the fewest syndromes uncovered, namely 11. The search starts from that 50-column set.

3.2 State and energy

A state is a set of exactly 50 distinct nonzero columns. An array of 1024 counters records, for each syndrome, its number of representations among the $1 + 50 + \binom{50}{2} = 1276$ sums of at most two columns. The energy of a state is its number of zero counters. A witness is a state of energy 0.

One column swap is cheap. Removal of a column c decrements the counter of c and the 49 counters of $c \oplus c'$ for the retained columns c'. Insertion increments the same pattern. A swap therefore costs about 100 counter updates instead of a 1276-term recount, and a rejected proposal is undone the same way.

3.3 Proposals and acceptance

With probability 220/256 a proposal targets a hole. The search picks an uncovered syndrome g uniformly. With probability 1/64 it offers the column g itself, which would cover g as a singleton. Otherwise it offers $g \oplus s$ for a uniform current column s, which would cover g as a pair through s. It draws at most 100 times for an offer that is nonzero and not already a column. If all draws fail, and in the remaining 36/256 of proposals, the offer is a uniform random non-member. The deleted column is uniform among the 50.

Acceptance is the Metropolis rule on the energy change Δ. Moves with $\Delta \leq 0$ always pass. Moves with $\Delta > 0$ pass with probability $e^{-\Delta/T}$.

Remark 2 (schedule, seed, replay). The temperature runs in cycles of 80,000 proposals. Within a cycle, $T = 3.5 \cdot 0.015^{\phi}$ as ϕ increases from 0 to 1, so each cycle cools from 3.5 to about 0.05. At the end of a cycle the state resets to the best state seen so far. The random source is xorshift64 with shift constants 13, 7, 17 and a fixed seed constant, called run 0. Run 0 reached energy 0 at proposal 3,600,281, which is 281 proposals after its 45th reset. The witness keeps 35 of the 50 seed columns. The command `python3 compute/search_radius2.py` verifies the committed witness with the standard library alone. The same command with `--search` replays run 0 and needs NumPy and Numba.

3.4 What the search did not close

The same method failed on three nearby gaps.

- At $(r, n) = (8, 25)$ the best anneal left 3 of 256 syndromes uncovered. Bounded runs of CP-SAT and Z3 ended with no verdict.

[1] The search ran unattended on 16 August 2026. The committed script `compute/search_radius2.py` is a deterministic reproduction of that run and fixes every constant quoted in this section.

- At $(9,38)$ a 300-second CP-SAT run returned `UNKNOWN`, and every single deletion from a reconstructed Gabidulin 39-set left at least 8 of 512 syndromes uncovered.
- At $(10,49)$ the anneal stalls at 7 of 1024.

These incomplete searches are not lower bounds.

Later work measured the depth of the $(10,49)$ incomplete search (`ATTACK.md`, 2026-08-18 and 19). Fresh deterministic anneals stall at 7 holes again. The best 7-hole state admits no swap of at most 5 columns that reaches 0 holes. That claim is exhaustive over all $\binom{49}{5} = 1{,}906{,}884$ removal sets with an exact re-insertion search, validated on four of four planted controls. A second 7-hole optimum admits no swap of at most 4 columns, and a backtracking search shows that no linear map of $\mathbb{F}_2^{10}$ carries one 49-set to the other. The landscape therefore has at least two distinct deep local optima at 7 holes. Separately, 79 subgroup classes of $GL(10,2)$ were exhausted with no invariant 49-covering. None of these facts bounds $\ell_2(10,2)$ from below. They say the known near-misses are far from any witness. Replay: `python3 compute/q4/verify_config.py compute/q4/best_sa_config.cols` prints the seven holes, and the exhaustion logs are `compute/q4/kswap5_sa.log` and `compute/q4/orbit_runs.log`.

3.5 Provenance of the partition

The search above produced the matrix only. A separate search on the same day produced the 10-block partition. That search code is not in the repository. The committed artifact is the partition itself, `result/data/partition_p10.json`, and the two independent verifiers of Section 4 accept it. Both verifiers re-derive the columns from the matrix text and enumerate all 1024 syndromes. The partition file supplies block labels only.

The pair statistics of Proposition 3 constrain any such search. Each of the 821 unique-pair syndromes forces its two columns into distinct blocks. These constraints form a graph with 821 edges on the 50 columns. A valid $(2,0)$-partition is a proper coloring of this graph that, for each of the remaining 152 syndromes, also splits at least one of its 2 to 6 pairs. Most of the partition is therefore forced before any search begins, and verification of a candidate costs one pass over the 1276 sums. Ten blocks satisfies the hypothesis $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$ of (2) for both $m=4$ and $m=5$.

4 Verification

4.1 Replay

Clone the public repository and run the pipeline:

```
git clone https://github.com/wustep/maths.git
cd maths/problems/covering/result
./run_all.sh
```

The pipeline needs `python3` and `rustc`. After the clone it uses no network, no packages, and no randomness. It executes 74 named assertions, exits with status 0, and prints byte-identical output across runs. A run takes about two seconds. Direct links: `https://github.com/wustep/maths/tree/main/problems/covering/result` and `result/run_all.sh`.

4.2 Two verifiers with opposite algorithms

`verify/verify.py` is pair-driven. It enumerates all $\binom{n}{2}$ pairs, marks the syndromes they reach, then sweeps all 2^r values for unmarked ones. `verify/verify.rs` is syndrome-driven. For each of the 2^r syndromes s it scans the columns h and tests membership of $s \oplus h$. It then runs its own pair loop and reports only if its two verdicts agree. The two programs also differ in parser and in rank pivot choice, lowest versus highest set bit, and both use exact rational arithmetic. `run_all.sh` compares their fact dumps line by line.

Both programs read the matrix from text. Neither reads a stored certificate.

4.3 Encoding

Every file in the repository is LSB-first: bit i of a column integer is row $i+1$. The hex listing of M_{KR} in Theorem 4.3 of [1] is MSB-first, with row 1 as the high bit. Reconstruction of $\boldsymbol{H}_{KR} = [\, I_{10} \mid M_{KR} \,]$ therefore reverses the ten bits of each column. A verifier that misses the reversal fails on the Kaikkonen–Rosendahl baseline and only there, which makes the mistake visible. The builder asserts the relation $h_5 + h_{27} + h_{29} = 0$ of Theorem 5.2(ii) of [1] as a guard.

A reader who uses the MSB-first convention obtains the bit-reversal of every column. Bit reversal is an invertible linear map on $\mathbb{F}_2^{10}$, so every claim in this note is preserved.

4.4 Scope of independence

One author wrote both verifiers in one session. The programs share no code and differ in language, algorithm, and parser, but they are not the work of two people. A third program, kept outside the repository, ran once after both verifiers agreed and confirmed every value.

5 What is not claimed

- Optimality. The sphere-covering bound gives $\ell_2(10,2) \geq 45$, since $1+n+\binom{n}{2} \geq 1024$ fails at $n = 44$ with 991. Proposition 2 concerns this 50-set only. A covering 49-set may exist.
- The $(10,49)$ runs that leave 7 syndromes uncovered are an incomplete search, not a lower bound. The leftover is deep (Section 3), but the depth of an incomplete search is still not a bound.
- The gaps $(8,25)$ and $(9,38)$ remain open.
- Only the $m = 4$ and $m = 5$ outputs are verified exhaustively. The later rows of the family inherit $p(\boldsymbol{H}_C) \leq 2^{m+1} + 1$ from (4.4) of [1].
- Within `result/`, nothing is claimed for $r \in \{12, 14, 16, 22, 24, 26, 28, 42, 44\}$. Later work certified $r = 22, 24, 26, 28$ from this seed (Remark 1). The values $r = 12, 14, 16$ carry no claim from this seed.
- $f(2) \leq 2601/2048$ is an upper bound only. Whether $f(2) = 1$ is open.
- Priority is unresolved. Table 5.1 of [1] is "best as far as the authors know". A proof of $\ell_2(10,2) \leq 50$ may exist in older proceedings. See `result/PRIORITY.md` in the repository.

6 Files

Everything named below is in the public repository `https://github.com/wustep/maths`, under `problems/covering/` on `main`.

in the repo	what
`result/NOTE.md`	repo companion
`result/note.tex`	arXiv-style draft of the same claims
`result/run_all.sh`	the verification pipeline
`result/data/H_r10_n50.txt`	the matrix
`result/data/partition_p10.json`	the 10-block partition
`compute/search_radius2.py`	the search, deterministic replay
`explainer.html`	interactive companion
`WALKTHROUGH.md`	narrative log of the discovery run
`ATTACK.md`	chronological log

The HTML companion is `explainer.html`. The matrix is `H_r10_n50.txt`.

A The matrix as bits

Rows 1 to 10 from top to bottom, columns left to right. Bit i (LSB) of each integer column is row $i+1$.

```
10010010011001010110111110011010010001111011111110
0101000100110100000010011101111100011110001100001010
00110001011011000011111011101100111000111110101101
00010000010100010101010001110100110001001000101011
00001001001011000010110110101000010100110101101100
00000100011001010001110010011101001100010011100011
00000011000111001111110001111011011100001111100111
00000000111111000000001111111000111100000000011111
00000000000000111111111111111000000011111111111111
00000000000000000000000000000111111111111111111111
```

References

[1] A. A. Davydov, S. Marcugini, F. Pambianco, *New upper bounds for binary linear covering codes*, arXiv:2511.02542 [cs.IT], November 2025.

[2] M. Kaikkonen, C. Rosendahl, *New covering codes from an ADS-like construction*, IEEE Trans. Inform. Theory **49** (2003), no. 7, 1809–1812.

[3] G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.

[4] B. Green, *100 open problems*, Problem 40. `https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf`.